But most notably, the confirmation of supersymmetry, through the discovery of superpartner particles as discussed in Chapter 9, would be a major milestone for string theory. We recall that supersymmetry was discovered in the course of theoretical investigations of string theory, and that it is a central part of the theory. Its experimental confirmation would be a compelling, albeit circumstantial, piece of evidence for strings. Moreover, finding the superpartner particles would provide a welcome challenge, since the discovery of supersymmetry would do far more than merely answer the yes-no question of its relevance to our world. The masses and charges of the superpartner particles would reveal the detailed way in which supersymmetry is incorporated into the laws of nature. String theorists would then face the challenge of seeing whether this implementation can be fully realized or explained by string theory. Of course, we can be even more optimistic and hope that within the next decade—before the Large Hadron Collider in Geneva comes on-line—the understanding of string theory will have progressed sufficiently for detailed predictions about the superpartners to be made prior to their hoped-for discovery. Confirmation of such predictions would be a monumental moment in the history of science.

Are There Limits to Explanation?

Explaining everything, even in the circumscribed sense of understanding all aspects of the forces and the elementary constituents of the universe, is one of the greatest challenges science has ever faced. And for the first time, superstring theory gives us a framework that appears to have sufficient depth to meet the challenge. But will we ever realize the promise of the theory fully and, for example, calculate the masses of the quarks or the strength of the electromagnetic force, numbers whose precise values dictate so much about the universe? As in the previous sections, we will have to surmount numerous theoretical hurdles on the way to these goals—currently, the most prominent is achieving a full nonperturbative formulation of string/M-theory.

But is it possible that even if we had an exact understanding of string/M-theory, framed within a new and far more transparent formulation of quantum mechanics, we could still fail in our quest to calculate particle masses and force strength? Is it possible that we would still have to resort to experimental measurements, rather than